Risk Monitoring and Control Effectiveness Validation Using File Integrity Monitoring (Wazuh Lab)

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Executive Summary

This work presents a practical implementation and validation of a **File Integrity Monitoring (FIM)** control using Wazuh, aligned with Governance, Risk, and Compliance (GRC) principles. The objective was to simulate a realistic risk scenario involving unauthorized changes to sensitive financial documents and to verify whether the implemented control can effectively detect, log, and report such activities.

A controlled environment was established using a monitored directory (C:\SensitiveFiles) on a Windows endpoint. Three critical risk events were simulated using Command Prompt: file creation, file modification, and file deletion. These actions represent common real-world threats such as insider misuse, unauthorized access, or malicious data manipulation.

The Wazuh platform successfully detected all simulated events in real time. Each activity was logged with detailed evidence, including file path, timestamp, event type, and cryptographic hash changes. This confirms that the control is functioning as a **detective control**, capable of identifying integrity violations as they occur.

Beyond detection, Wazuh enriched each alert with mappings to major compliance frameworks such as **PCI DSS, HIPAA, NIST 800-53, GDPR, TSC, and MITRE ATT&CK**. This demonstrates that the control not only provides technical monitoring but also produces **audit-ready evidence**, reducing the burden of manual compliance mapping and supporting regulatory reporting.

From a GRC perspective, the implementation shows how technical controls can be directly tied to risk management objectives. The generated alerts enable risk quantification through measurable metrics such as event frequency, severity levels, and affected assets. The centralized dashboard further enhances governance by improving visibility, accountability, and auditability across the environment.

In summary, this work confirms that Wazuh FIM is an effective control for detecting unauthorized file activities, supporting compliance requirements, and strengthening organizational risk monitoring capabilities.

Lab Environment Setup

1. Overview

This section documents the complete setup of the laboratory environment used to implement and validate a detective security control using Wazuh File Integrity Monitoring (FIM).

The objective of this environment is to simulate a real-world enterprise monitoring architecture where a centralized security platform collects, analyzes, and reports endpoint activity for risk monitoring and audit purposes.

The environment consists of:

- A Wazuh Manager deployed on Ubuntu (host system)
- A Windows 10 virtual machine acting as a monitored endpoint
- A Wazuh Agent installed on the Windows system
- Network connectivity between both systems for log forwarding and control validation

2. Architecture Design

The lab follows a centralized monitoring architecture:

Component	Role
Ubuntu Host	Wazuh Manager, Indexer, Dashboard
Windows VM	Monitored endpoint
Wazuh Agent	Endpoint data collection
Wazuh Dashboard	Visualization and analysis

All endpoint activities are forwarded from the Windows agent to the Wazuh manager for analysis and alert generation.

3. Wazuh Server Deployment (Ubuntu)

The Wazuh platform was installed using the official automated installation script, which deploys:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard
- Filebeat service

Verification of Services

All core services were verified to be active and running.

Wazuh Manager Status

```
amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~
● wazuh-manager.service - Wazuh manager
     Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-04-30 08:26:01 WAT; 1h 50min ago
    Process: 2415 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=>
      Tasks: 174 (limit: 9201)
     Memory: 240.3M (peak: 1.4G swap: 217.1M swap peak: 283.2M)
        CPU: 2min 40.325s
     CGroup: /system.slice/wazuh-manager.service
             ├─2878 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─2879 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─2880 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─2883 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─2886 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─2928 /var/ossec/bin/wazuh-authd
             ├─2943 /var/ossec/bin/wazuh-db
             ├─2969 /var/ossec/bin/wazuh-execd
             ├─2983 /var/ossec/bin/wazuh-analysisd
             ├─2992 /var/ossec/bin/wazuh-syscheckd
             ├─3009 /var/ossec/bin/wazuh-remoted
             ├─3094 /var/ossec/bin/wazuh-logcollector
             ├─3115 /var/ossec/bin/wazuh-monitord
             └─3138 /var/ossec/bin/wazuh-modulesd

Apr 30 08:25:54 ayuba env[2415]: Started wazuh-analysisd...
Apr 30 08:25:55 ayuba env[2415]: Started wazuh-syscheckd...
```

Wazuh Indexer Status

```
amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~
● wazuh-indexer.service - wazuh-indexer
     Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-04-30 08:26:14 WAT; 1h 51min ago
       Docs: https://documentation.wazuh.com
   Main PID: 2414 (java)
      Tasks: 83 (limit: 9201)
     Memory: 905.7M (peak: 1.5G swap: 589.1M swap peak: 1.1G)
        CPU: 3min 19.150s
     CGroup: /system.slice/wazuh-indexer.service
             └─2414 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddres>

Apr 30 08:25:45 ayuba systemd-entrypoint[2414]: WARNING: System::setSecurityManager has been call>
Apr 30 08:25:45 ayuba systemd-entrypoint[2414]: WARNING: Please consider reporting this to the ma>
Apr 30 08:25:45 ayuba systemd-entrypoint[2414]: WARNING: System::setSecurityManager will be remov>
Apr 30 08:25:46 ayuba systemd-entrypoint[2414]: Apr 30, 2026 8:25:46 AM sun.util.locale.provider.>
Apr 30 08:25:46 ayuba systemd-entrypoint[2414]: WARNING: COMPAT locale provider will be removed i>
Apr 30 08:25:48 ayuba systemd-entrypoint[2414]: WARNING: A terminally deprecated method in java.l>
Apr 30 08:25:48 ayuba systemd-entrypoint[2414]: WARNING: System::setSecurityManager has been call>
Apr 30 08:25:48 ayuba systemd-entrypoint[2414]: WARNING: Please consider reporting this to the ma>
Apr 30 08:25:48 ayuba systemd-entrypoint[2414]: WARNING: System::setSecurityManager will be remov>
Apr 30 08:26:14 ayuba systemd[1]: Started wazuh-indexer.service - wazuh-indexer.
```

Filebeat Status

```
amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~
amadi@ayuba:~$ sudo systemctl status filebeat
[sudo] password for amadi:
● filebeat.service - Filebeat sends log files to Logstash or directly to E>
     Loaded: loaded (/usr/lib/systemd/system/filebeat.service; enabled; pr>
     Active: active (running) since Thu 2026-04-30 08:25:40 WAT; 1h 52min >
       Docs: https://www.elastic.co/products/beats/filebeat
   Main PID: 2410 (filebeat)
      Tasks: 10 (limit: 9201)
     Memory: 16.9M (peak: 49.5M swap: 2.2M swap peak: 2.5M)
        CPU: 2.190s
     CGroup: /system.slice/filebeat.service
             └─2410 /usr/share/filebeat/bin/filebeat --environment systemd>

Apr 30 08:25:40 ayuba systemd[1]: Started filebeat.service - Filebeat send>
lines 1-12/12 (END)
```

Wazuh Dashboard Service Status

```
amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~   amadi@ayuba: ~
● wazuh-dashboard.service - wazuh-dashboard
     Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-04-30 08:25:30 WAT; 1h 52min ago
   Main PID: 1026 (node)
      Tasks: 11 (limit: 9201)
     Memory: 149.1M (peak: 384.5M swap: 58.1M swap peak: 154.0M)
        CPU: 32.573s
     CGroup: /system.slice/wazuh-dashboard.service
             └─1026 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size>

Apr 30 08:30:00 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T07:30:0>
Apr 30 08:45:03 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T07:45:0>
Apr 30 09:00:01 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T08:00:0>
Apr 30 09:15:00 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T08:15:0>
Apr 30 09:30:01 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T08:30:0>
Apr 30 09:45:01 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T08:45:0>
Apr 30 10:00:00 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T09:00:0>
Apr 30 10:00:00 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T09:00:0>
Apr 30 10:15:03 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T09:15:0>
Apr 30 10:15:03 ayuba opensearch-dashboards[1026]: {"type":"log","@timestamp":"2026-04-30T09:15:0>
~
~
```

4. Wazuh Dashboard Access

The Wazuh web interface was successfully accessed via:

https://10.10.10.227

The dashboard provides:

- Agent monitoring
- Alert visualization

- Threat detection modules
- File integrity monitoring insights

5. Windows Endpoint Setup

A Windows 10 virtual machine was deployed using VirtualBox to simulate an enterprise endpoint.

Network Configuration

- Adapter Type: Bridged Adapter
- Network Range: 10.10.10.0/24
- Windows IP Address: 10.10.10.70
- Ubuntu (Wazuh Manager) IP Address: 10.10.10.227

Connectivity between systems was validated using ICMP (ping), confirming successful communication.

6. Wazuh Agent Installation and Registration

The Wazuh agent was installed on the Windows VM and configured to communicate with the Wazuh manager.

Agent Registration Process

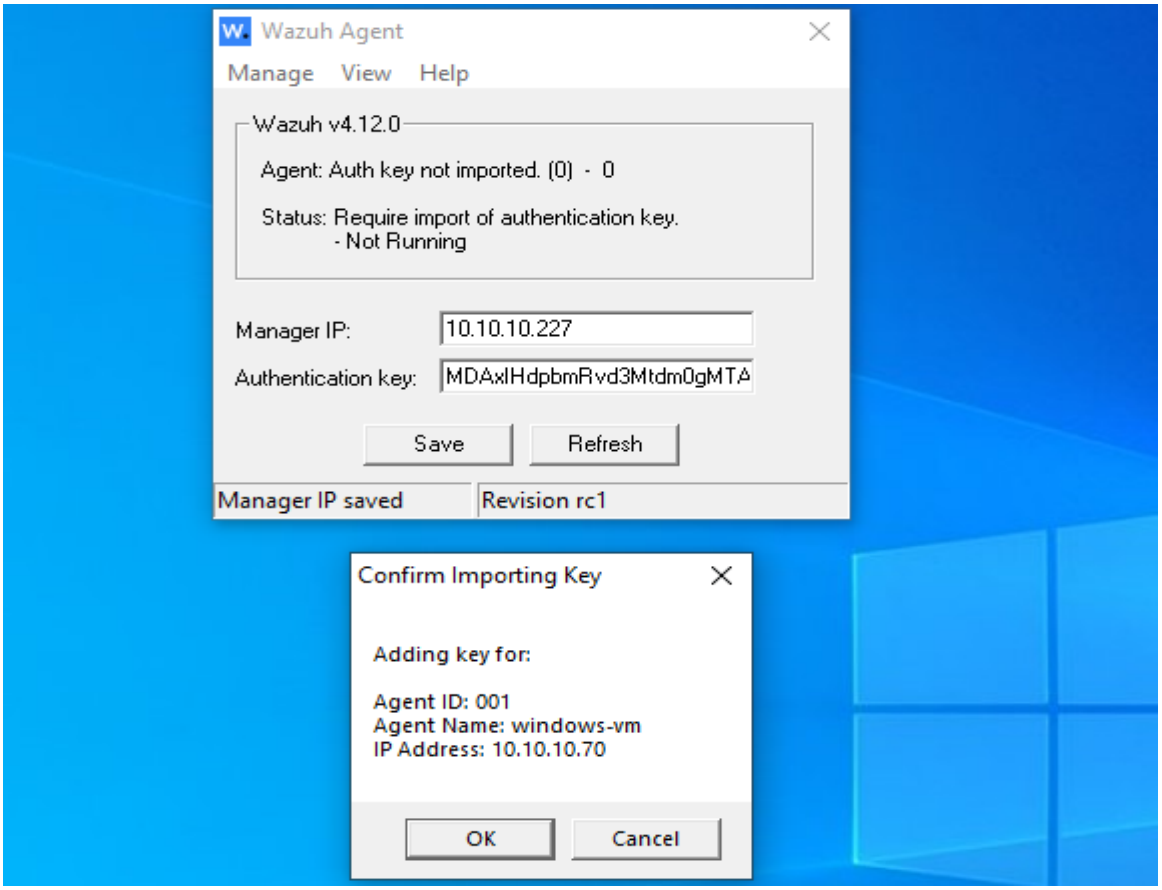
The agent was manually registered on the Wazuh manager using the agent management utility.

Agent Creation and Key Extraction

```
amadi@ayuba:~$ sudo /var/ossec/bin/manage_agents
[sudo] password for amadi:


****************************************
* Wazuh v4.12.0 Agent manager.         *
* The following options are available: *
****************************************
   (A)dd an agent (A).
   (E)xtract key for an agent (E).
   (L)ist already added agents (L).
   (R)emove an agent (R).
   (Q)uit.
Choose your action: A,E,L,R or Q: A

- Adding a new agent (use '\q' to return to the main menu).
  Please provide the following:
   * A name for the new agent: windows-vm
   * The IP Address of the new agent: 10.10.10.70
Confirm adding it?(y/n): Y
Agent added with ID 001.


****************************************
* Wazuh v4.12.0 Agent manager.         *
* The following options are available: *
****************************************
   (A)dd an agent (A).
   (E)xtract key for an agent (E).
   (L)ist already added agents (L).
   (R)emove an agent (R).
   (Q)uit.
Choose your action: A,E,L,R or Q: E

Available agents:
   ID: 001, Name: windows-vm, IP: 10.10.10.70
Provide the ID of the agent to extract the key (or '\q' to quit): 01

Agent key information for '001' is:
MDAxIHdpbmRvd3Mtdm0gMTAuMTAuMTAuNzAgM2I0YTIwMThlZjE2MGVhNzg4OWQ2NzllM2RlMjIyNDYyM2NlNTVhMWIxMzNmNjc2YmIyOTAzNDI0Zjc2Nj
Y4Nw==

** Press ENTER to return to the main menu.
```

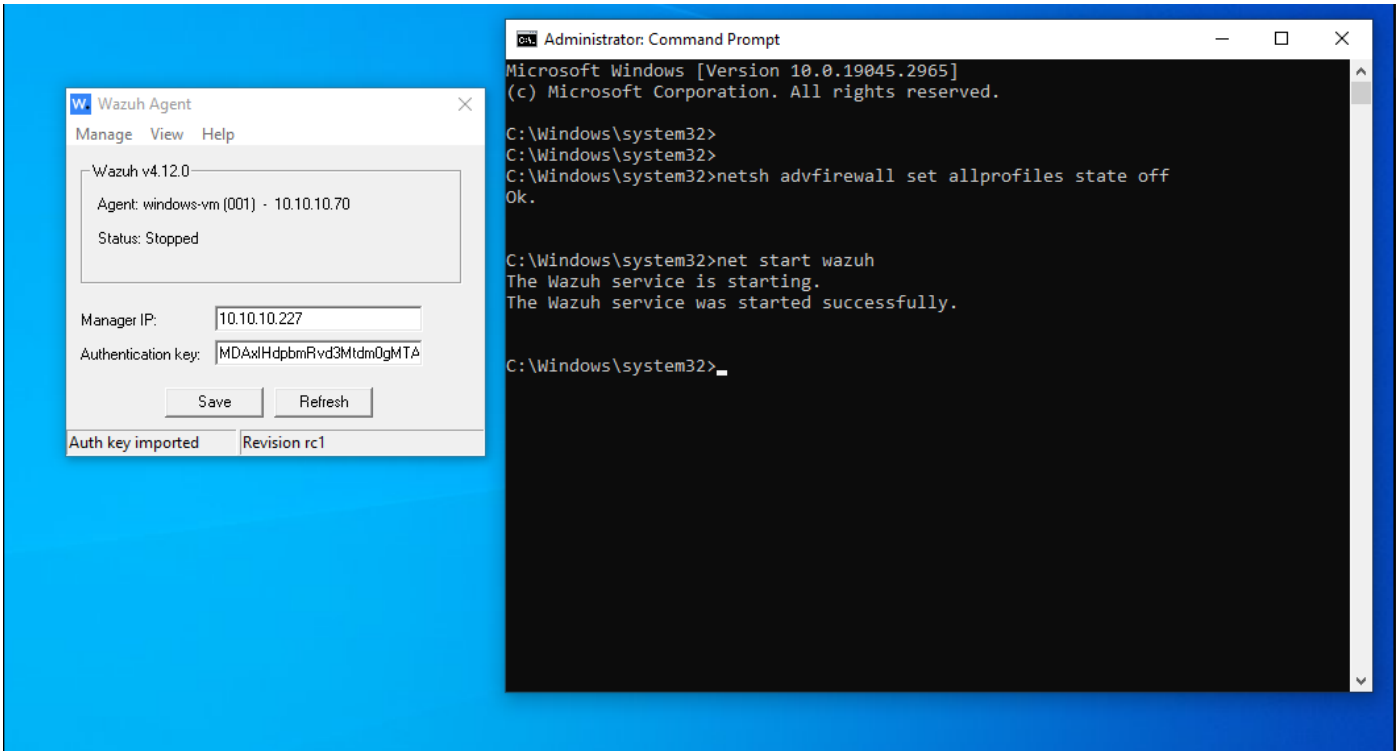
Agent Key Import on Windows

The extracted authentication key was imported into the Windows agent interface.



Agent Activation

The Wazuh agent service was started successfully on the Windows system.



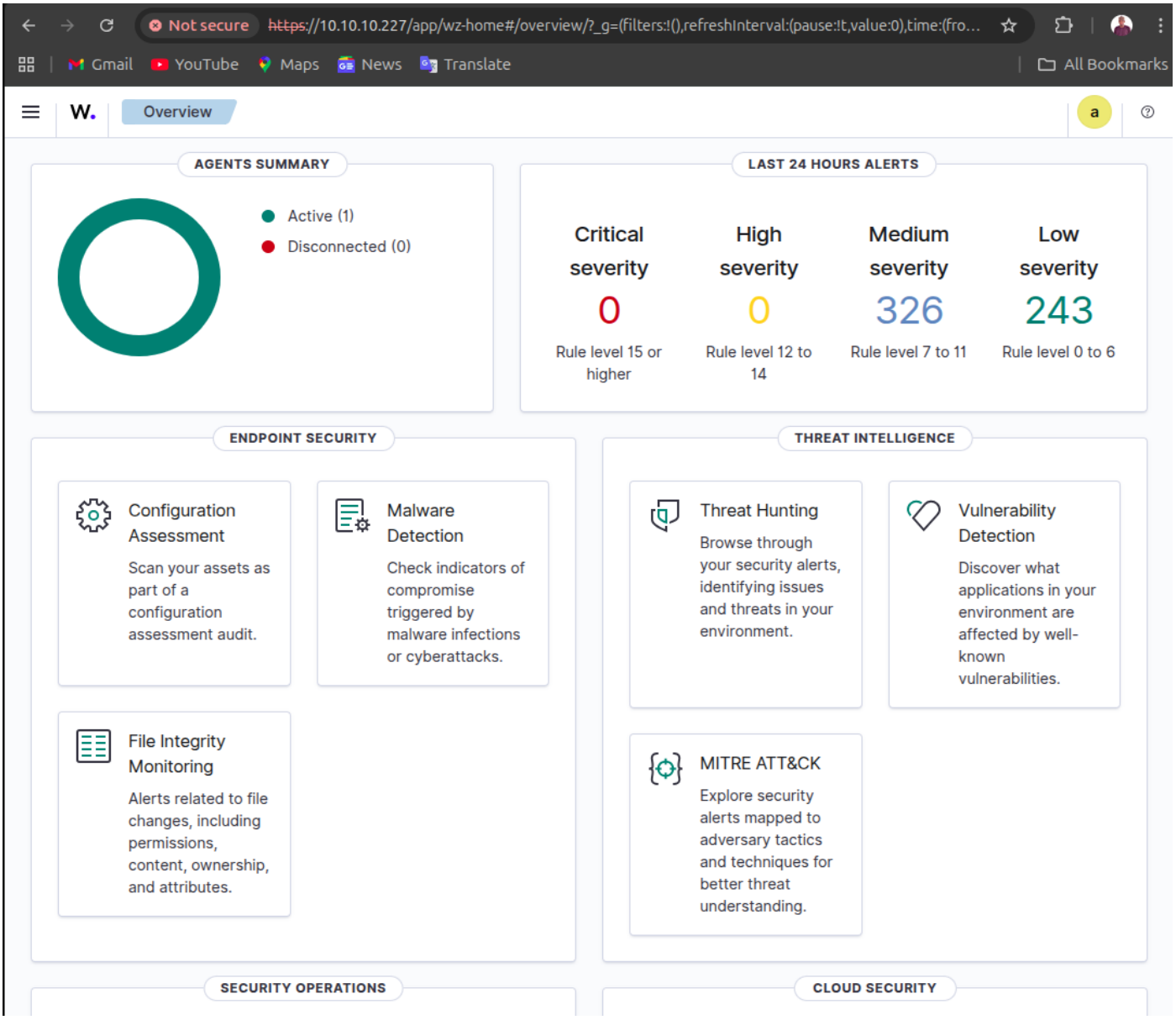
7. Agent Connectivity Verification

After configuration, the agent successfully connected to the Wazuh manager.

This confirms:

- Proper network configuration
- Successful authentication
- Active data transmission

Agent Status in Dashboard



8. File Integrity Monitoring Configuration

To support the lab objective, File Integrity Monitoring (FIM) was configured on the Windows agent.

The configuration file:

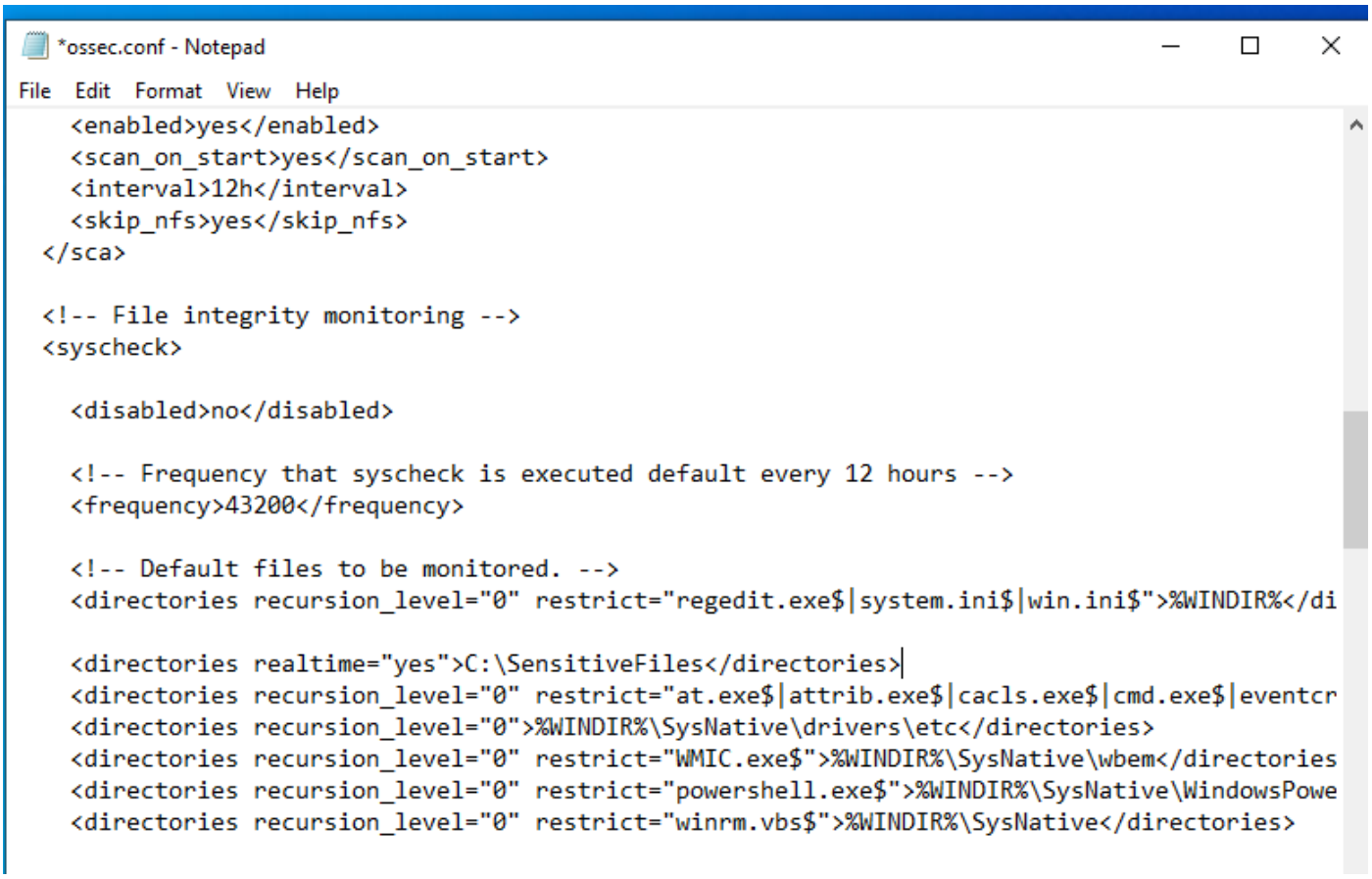
C:\Program Files (x86)\ossec-agent\ossec.conf

was modified to include the monitored directory:

<directories realtime="yes">C:\SensitiveFiles</directories>

This enables real-time monitoring of file activity within the specified directory.

Configuration File Update



Risk Monitoring and Control Verification

1. Objective

The objective of this phase was to verify that the File Integrity Monitoring control implemented through Wazuh was operating effectively.

The specific risk being tested was the unauthorized creation, modification, or deletion of sensitive financial forecast documents. This aligns with the lab scenario, where unauthorized changes to financial documents were identified as a medium-to-high organizational risk requiring a detective control.

For this lab, the monitored directory was configured as:

C:\SensitiveFiles

This folder was used as the controlled location for testing file integrity monitoring.

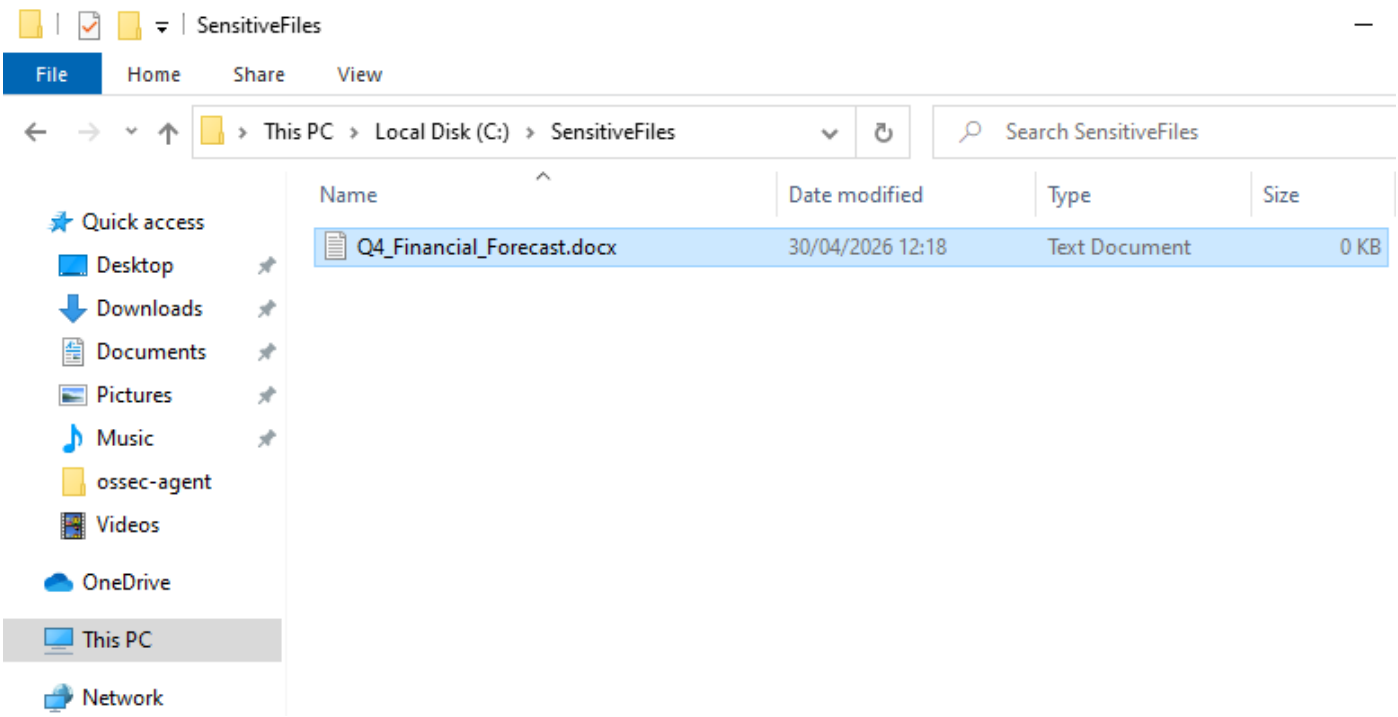
2. Monitored Folder Creation

A monitored folder was created on the Windows endpoint to represent the location where sensitive financial documents are stored.

C:\SensitiveFiles

The file used for testing was:

Q4_Financial_Forecast.docx



Explanation

This evidence shows the sensitive directory and the financial forecast file used for the monitoring test.

3. Risk Event Simulation

To test the detective control, three file activity events were simulated using Windows Command Prompt:

1. File creation

2. File modification
3. File deletion

These actions represent possible unauthorized activity against sensitive financial documents.

4. File Lifecycle Simulation Using CMD

The following commands were executed on the Windows endpoint.

File Creation

echo Q4 financial forecast document created for monitoring test > C:\SensitiveFiles\Q4_Financial_Forecast.docx

This command created the sensitive test file.

File Modification
echo Unauthorized update added to the financial forecast document >> C:\SensitiveFiles\Q4_Financial_Forecast.docx

This command modified the file by appending new content.

File Deletion
del C:\SensitiveFiles\Q4_Financial_Forecast.docx

This command deleted the file from the monitored directory.

```
C:\Users\Amadi's VM>echo Q4 financial forecast document created for monitoring test > C:\SensitiveFiles
\Q4_Financial_Forecast.docx

C:\Users\Amadi's VM>dir C:\SensitiveFiles
 Volume in drive C has no label.
 Volume Serial Number is 4850-D1F5

 Directory of C:\SensitiveFiles

30/04/2026  12:40    <DIR>          .
30/04/2026  12:40    <DIR>          ..
30/04/2026  12:40                61 Q4_Financial_Forecast.docx
               1 File(s)             61 bytes
               2 Dir(s)  28,461,019,136 bytes free

C:\Users\Amadi's VM>echo Unauthorized update added to the financial forecast document >> C:\SensitiveFi
les\Q4_Financial_Forecast.docx

C:\Users\Amadi's VM>type C:\SensitiveFiles\Q4_Financial_Forecast.docx
Q4 financial forecast document created for monitoring test
Unauthorized update added to the financial forecast document

C:\Users\Amadi's VM>del C:\SensitiveFiles\Q4_Financial_Forecast.docx

C:\Users\Amadi's VM>dir C:\SensitiveFiles
 Volume in drive C has no label.
 Volume Serial Number is 4850-D1F5

 Directory of C:\SensitiveFiles

30/04/2026  12:41    <DIR>          .
30/04/2026  12:41    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  28,460,892,160 bytes free

C:\Users\Amadi's VM>
```

This visual confirms that the file was created, modified, verified using the type of command, deleted, and then confirmed as removed from the monitored folder.

5. Monitoring Control Verification in Wazuh

After the file activity was performed, the Wazuh dashboard was used to verify that the File Integrity Monitoring control detected the events.

The following filter was applied in the Wazuh dashboard:

rule.groups: syscheck

This filter displays File Integrity Monitoring alerts only.

The dashboard confirmed that Wazuh detected:

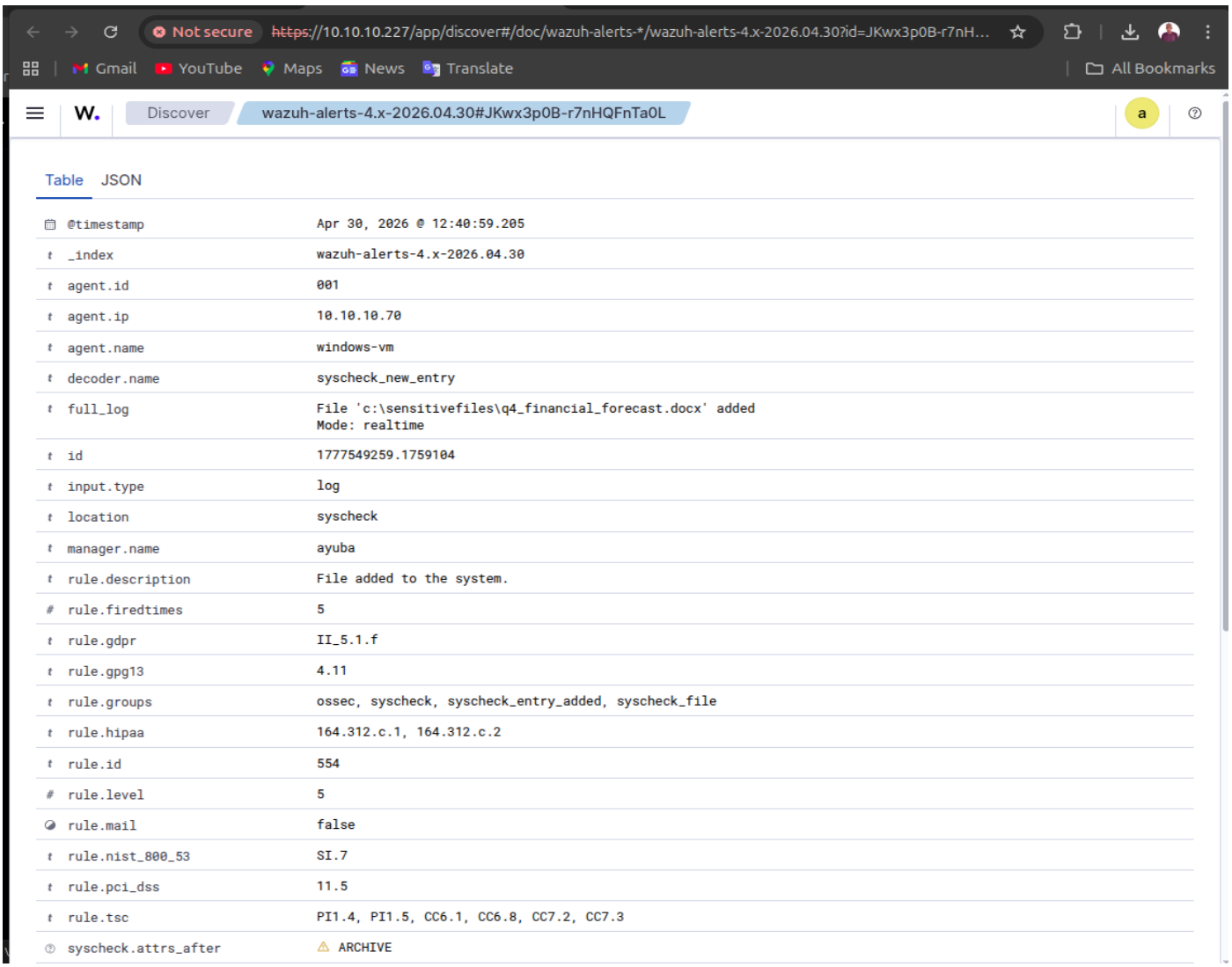
Event	Wazuh Detection
File creation	File added to the system
File modification	Integrity checksum changed
File deletion	File deleted

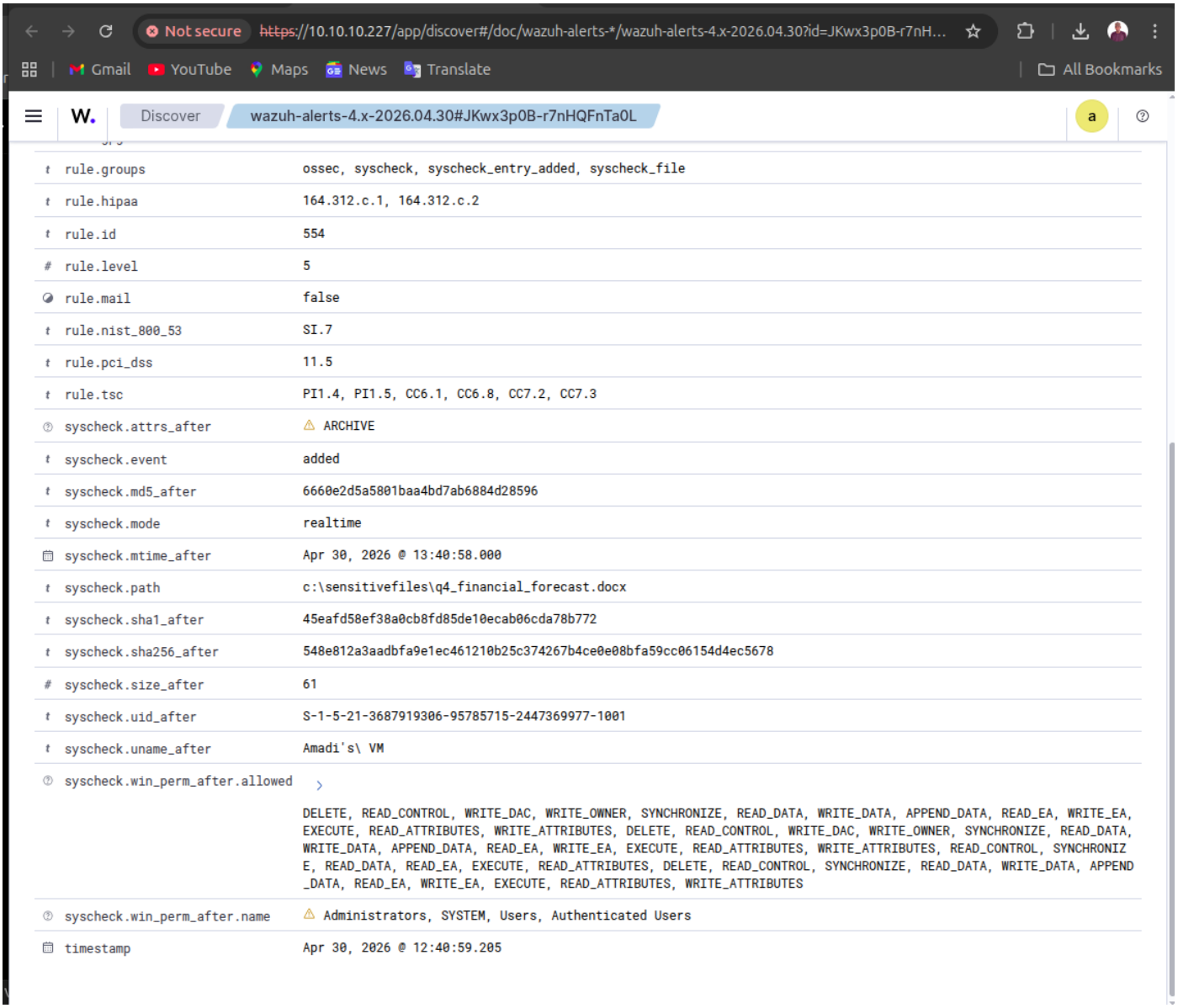
6. File Creation Alert

Wazuh detected the creation of the file inside the monitored directory.

Key Alert Details

Field	Value
Event type	File added
Rule description	File added to the system
File path	C:\SensitiveFiles\Q4_Financial_Forecast.docx
Agent	windows-vm
Detection mode	Realtime
Rule group	syscheck





rule.groups	ossec, syscheck, syscheck_entry_added, syscheck_file
rule.hipaa	164.312.c.1, 164.312.c.2
rule.id	554
rule.level	5
rule.mail	false
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
syscheck.attrs_after	ARCHIVE
syscheck.event	added
syscheck.md5_after	6660e2d5a5801baa4bd7ab6884d28596
syscheck.mode	realtime
syscheck.mtime_after	Apr 30, 2026 @ 13:40:58.000
syscheck.path	c:\sensitivefiles\q4_financial_forecast.docx
syscheck.sha1_after	45eafd58ef38a0cb8fd85de10ecab06cda78b772
syscheck.sha256_after	548e812a3aadbfa9e1ec461210b25c374267b4ce0e08bfa59cc06154d4ec5678
syscheck.size_after	61
syscheck.uid_after	S-1-5-21-3687919306-95785715-2447369977-1001
syscheck.uname_after	Amadi's\ VM
syscheck.win_perm_after.allowed	DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, READ_CONTROL, SYNCHRONIZE, READ_DATA, READ_EA, EXECUTE, READ_ATTRIBUTES, DELETE, READ_CONTROL, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES
syscheck.win_perm_after.name	Administrators, SYSTEM, Users, Authenticated Users
timestamp	Apr 30, 2026 @ 12:40:59.205

This confirms that the control detected the creation of a new file within the monitored sensitive directory.

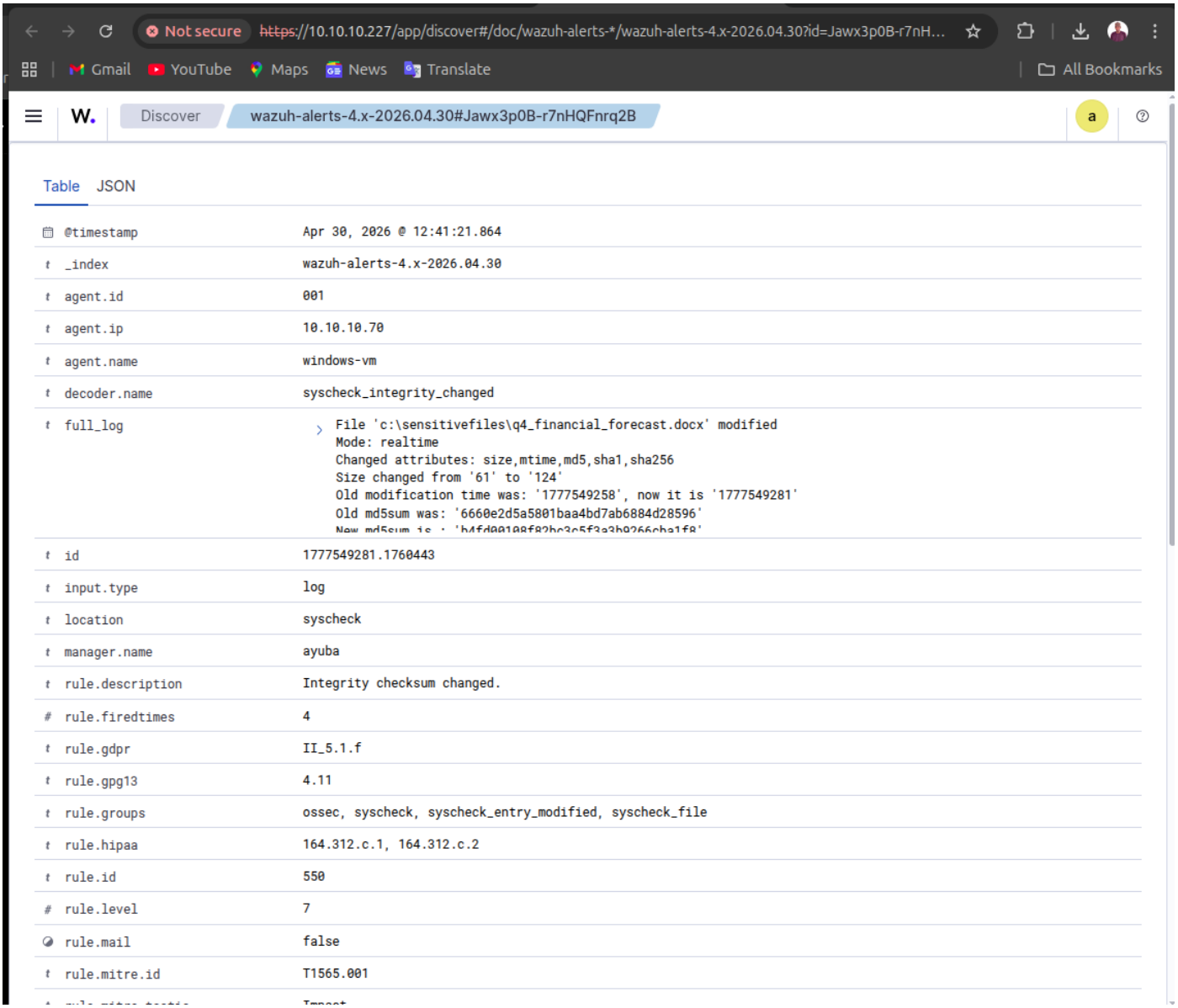
7. File Modification Alert

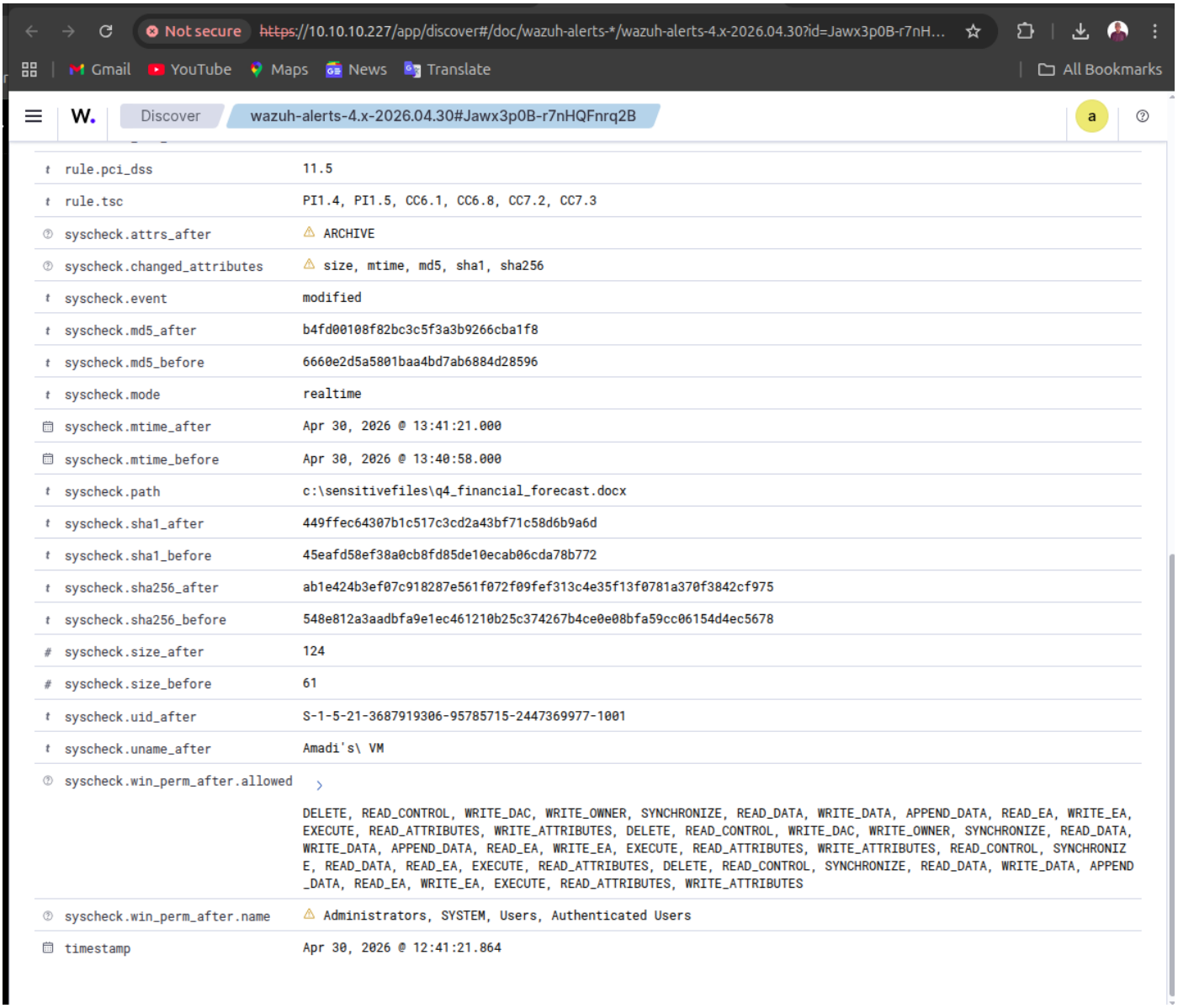
Wazuh detected that the file content changed.

Key Alert Details

Field	Value

Event type	Modified
Rule description	Integrity checksum changed
File path	C:\SensitiveFiles\Q4_Financial_Forecast.docx
Agent	windows-vm
Detection mode	Realtime
Rule group	syscheck
Size change	61 bytes to 124 bytes





Not secure https://10.10.10.227/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.04.30?id=Jawx3p0B-r7nH...

Gmail YouTube Maps News Translate All Bookmarks

W. Discover wazuh-alerts-4.x-2026.04.30#Jawx3p0B-r7nHQFnrq2B

rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
syscheck.attrs_after	ARCHIVE
syscheck.changed_attributes	size, mtime, md5, sha1, sha256
syscheck.event	modified
syscheck.md5_after	b4fd00108f82bc3c5f3a3b9266cba1f8
syscheck.md5_before	6660e2d5a5801baa4bd7ab6884d28596
syscheck.mode	realtime
syscheck.mtime_after	Apr 30, 2026 @ 13:41:21.000
syscheck.mtime_before	Apr 30, 2026 @ 13:40:58.000
syscheck.path	c:\sensitivefiles\q4_financial_forecast.docx
syscheck.sha1_after	449ffec64307b1c517c3cd2a43bf71c58d6b9a6d
syscheck.sha1_before	45eafd58ef38a0cb8fd85de10ecab06cda78b772
syscheck.sha256_after	ab1e424b3ef07c918287e561f072f09fef313c4e35f13f0781a370f3842cf975
syscheck.sha256_before	548e812a3aadbfa9e1ec461210b25c374267b4ce0e08bfa59cc06154d4ec5678
syscheck.size_after	124
syscheck.size_before	61
syscheck.uid_after	S-1-5-21-3687919306-95785715-2447369977-1001
syscheck.uname_after	Amadi's\ VM
syscheck.win_perm_after.allowed	DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, READ_CONTROL, SYNCHRONIZE, READ_DATA, READ_EA, EXECUTE, READ_ATTRIBUTES, DELETE, READ_CONTROL, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES
syscheck.win_perm_after.name	Administrators, SYSTEM, Users, Authenticated Users
timestamp	Apr 30, 2026 @ 12:41:21.864

This is strong evidence because Wazuh did not only detect that the file changed. It also recorded changes in file size and cryptographic hash values such as MD5, SHA1, and SHA256.

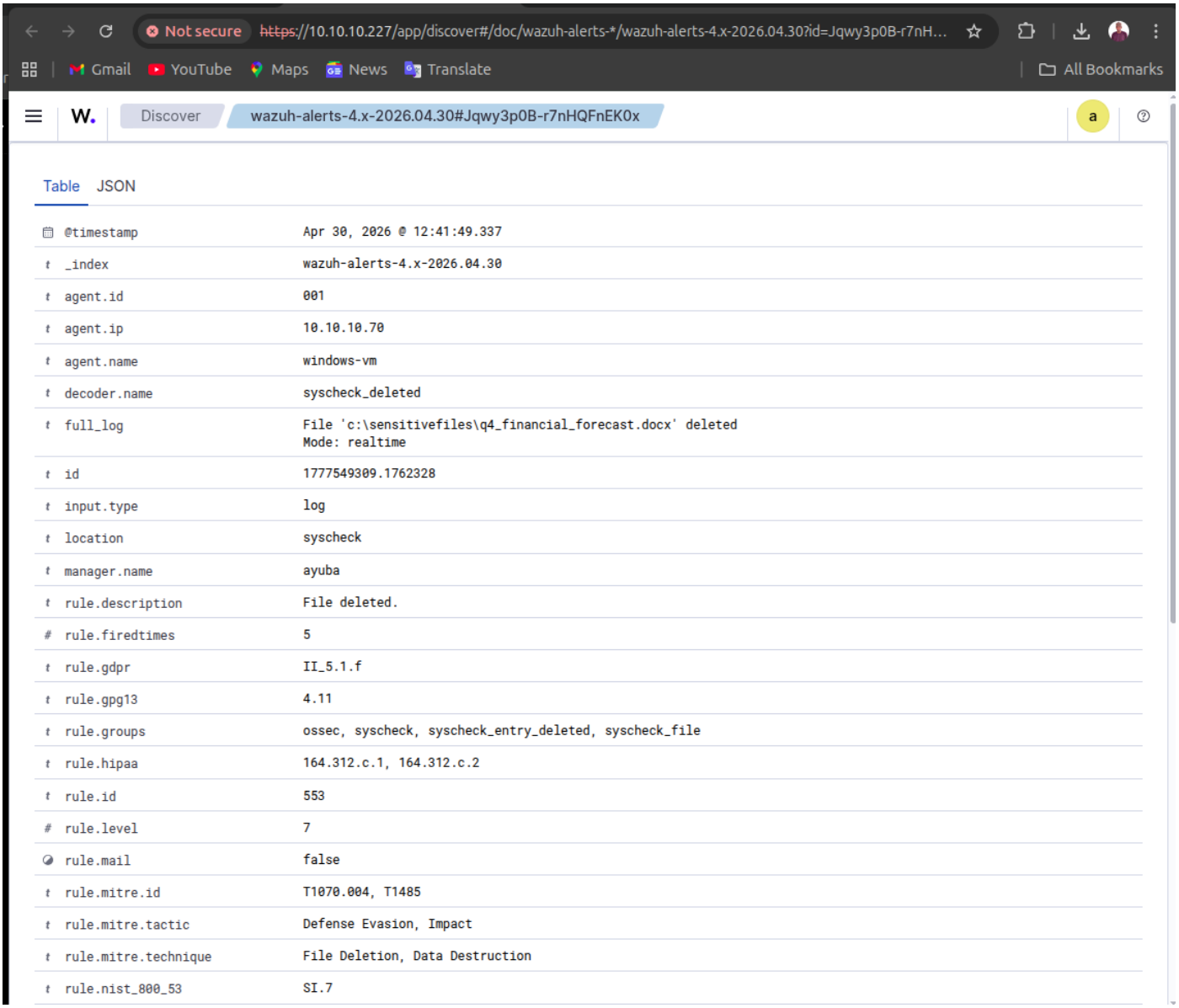
This proves that the file content was altered.

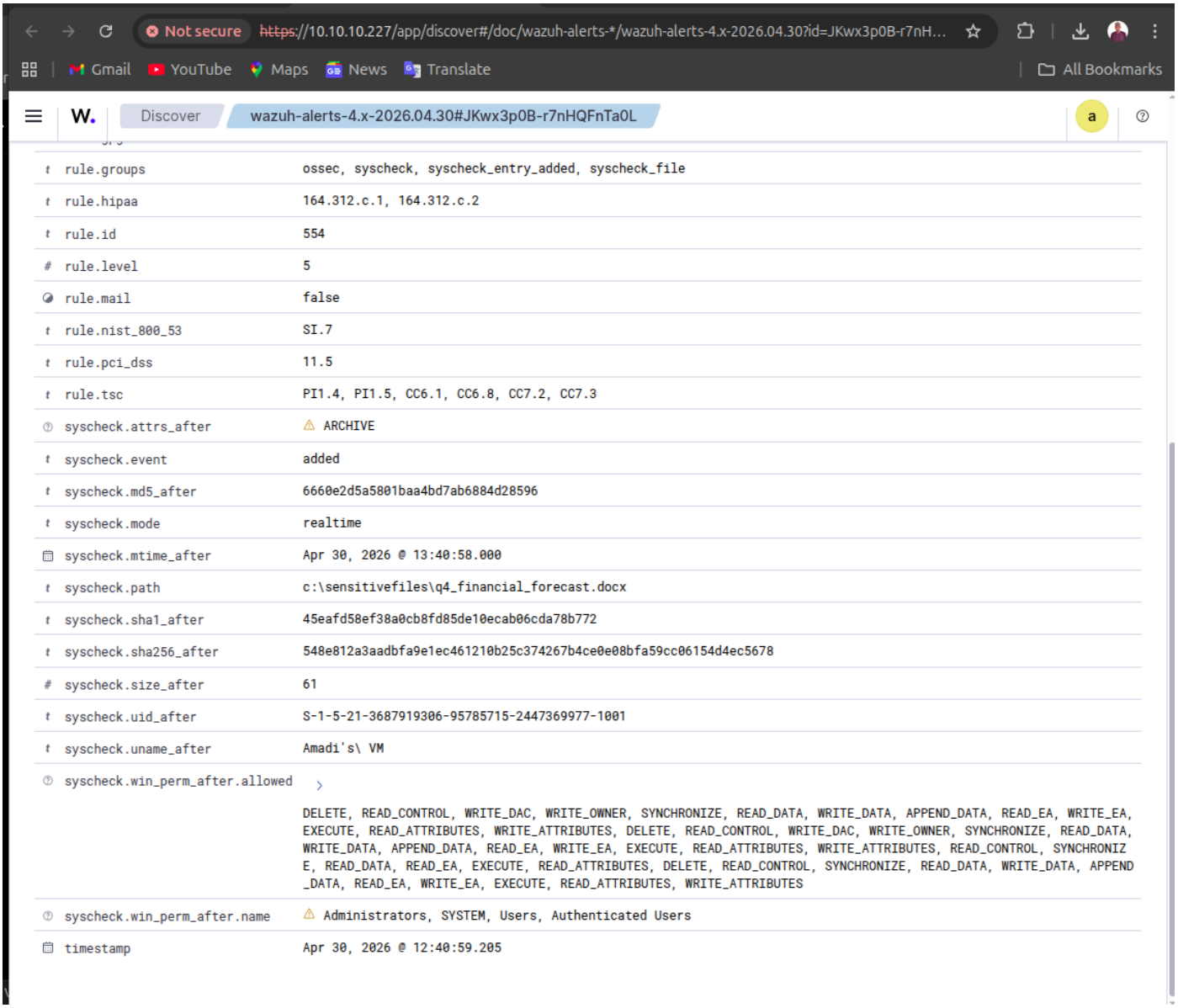
8. File Deletion Alert

Wazuh detected that the monitored file was deleted.

Key Alert Details

Field	Value
Event type	Deleted
Rule description	File deleted
File path	C:\SensitiveFiles\Q4_Financial_Forecast.docx
Agent	windows-vm
Detection mode	Realtime
Rule group	syscheck
MITRE mapping	File Deletion, Data Destruction





rule.groups	ossec, syscheck, syscheck_entry_added, syscheck_file
rule.hipaa	164.312.c.1, 164.312.c.2
rule.id	554
rule.level	5
rule.mail	false
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
syscheck.attrs_after	ARCHIVE
syscheck.event	added
syscheck.md5_after	6660e2d5a5801baa4bd7ab6884d28596
syscheck.mode	realtime
syscheck.mtime_after	Apr 30, 2026 @ 13:40:58.000
syscheck.path	c:\sensitivefiles\q4_financial_forecast.docx
syscheck.sha1_after	45eafd58ef38a0cb8fd85de10ecab06cda78b772
syscheck.sha256_after	548e812a3aadbfa9e1ec461210b25c374267b4ce0e08bfa59cc06154d4ec5678
syscheck.size_after	61
syscheck.uid_after	S-1-5-21-3687919306-95785715-2447369977-1001
syscheck.uname_after	Amadi's\ VM
syscheck.win_perm_after.allowed	DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, READ_CONTROL, SYNCHRONIZE, READ_DATA, READ_EA, EXECUTE, READ_ATTRIBUTES, DELETE, READ_CONTROL, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES
syscheck.win_perm_after.name	Administrators, SYSTEM, Users, Authenticated Users
timestamp	Apr 30, 2026 @ 12:40:59.205

This confirms that the control can detect deletion of sensitive files. In a real organization, this could indicate accidental deletion, insider misuse, or malicious data destruction.

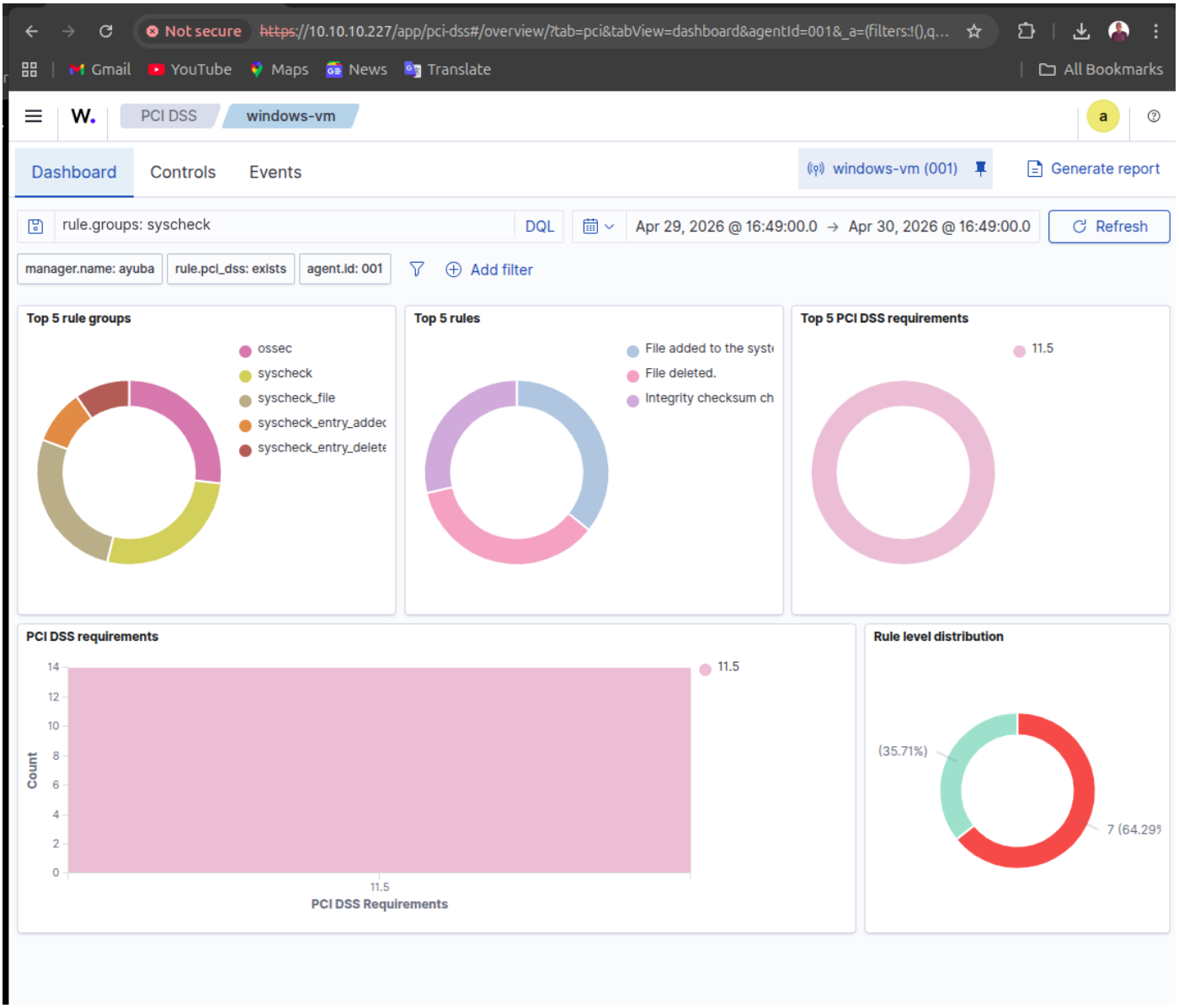
9. Compliance and Framework Mapping

Wazuh does not only generate technical alerts. It also enriches each alert with mappings to recognized compliance frameworks and security standards.

This is critical from a GRC perspective because it transforms raw security events into **audit-ready evidence**. Instead of manually mapping controls during an audit, the system already aligns detected activities with regulatory expectations.

In this lab, all File Integrity Monitoring (FIM) events generated from the monitored directory were automatically mapped to multiple frameworks, demonstrating that the control supports both **security monitoring** and **regulatory compliance**.

9.1 PCI DSS Mapping



Observed Mapping

Framework	Requirement
PCI DSS	11.5

PCI DSS Requirement 11.5 requires organizations to deploy file integrity monitoring mechanisms to detect unauthorized modification of critical files.

The alerts generated during this lab (file creation, modification, and deletion) directly satisfy this requirement. Wazuh recorded each event with timestamps, file paths, and integrity changes, proving that monitoring is active and effective.

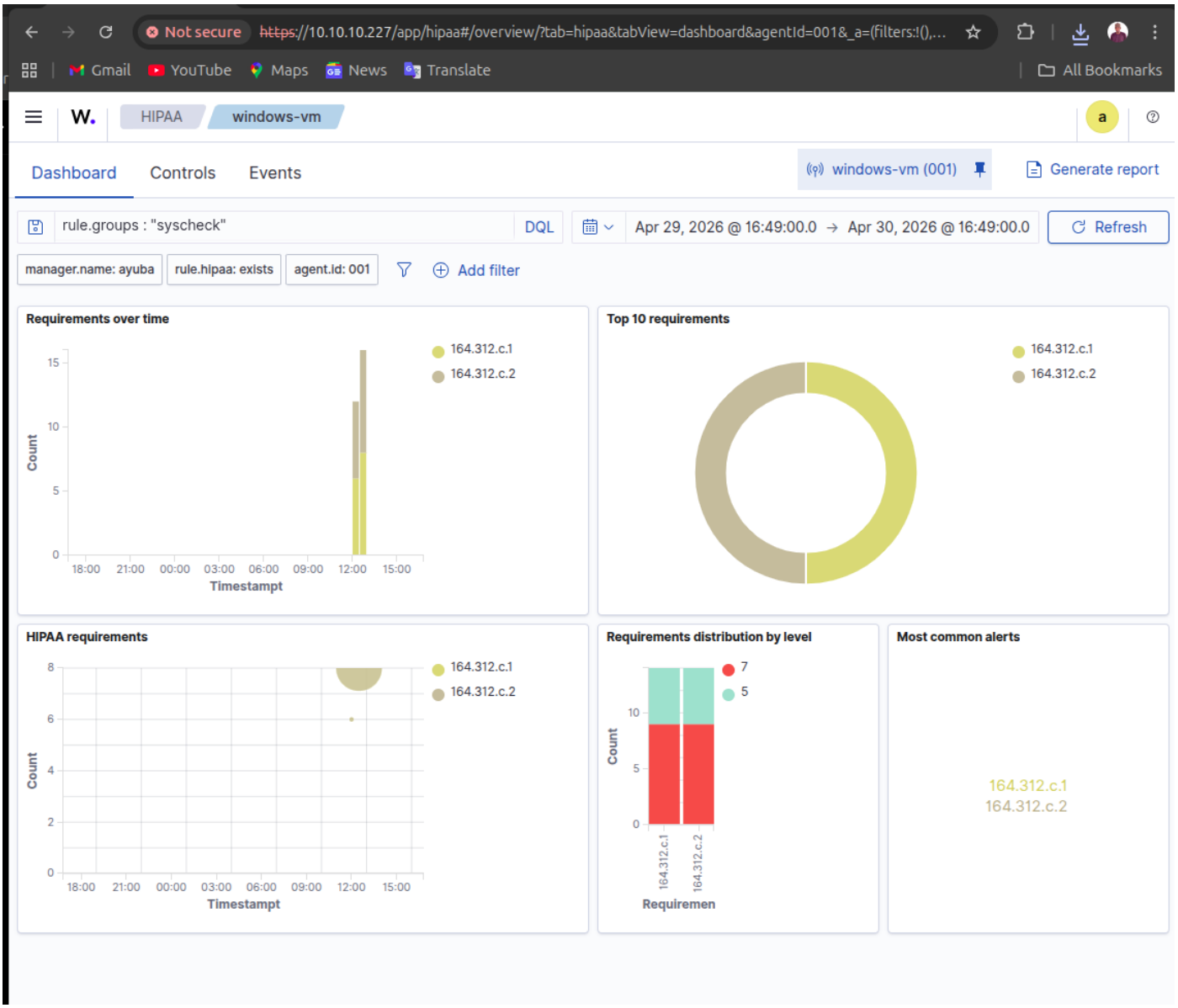
From an audit perspective, this demonstrates that:

- File integrity monitoring is implemented

- Changes are detected in real time
- Evidence is centrally logged and retrievable

This removes the need for manual proof during compliance assessments.

9.2 HIPAA Mapping



Observed Mapping

Framework	Requirement
HIPAA	164.312(c)(1)
HIPAA	164.312(c)(2)

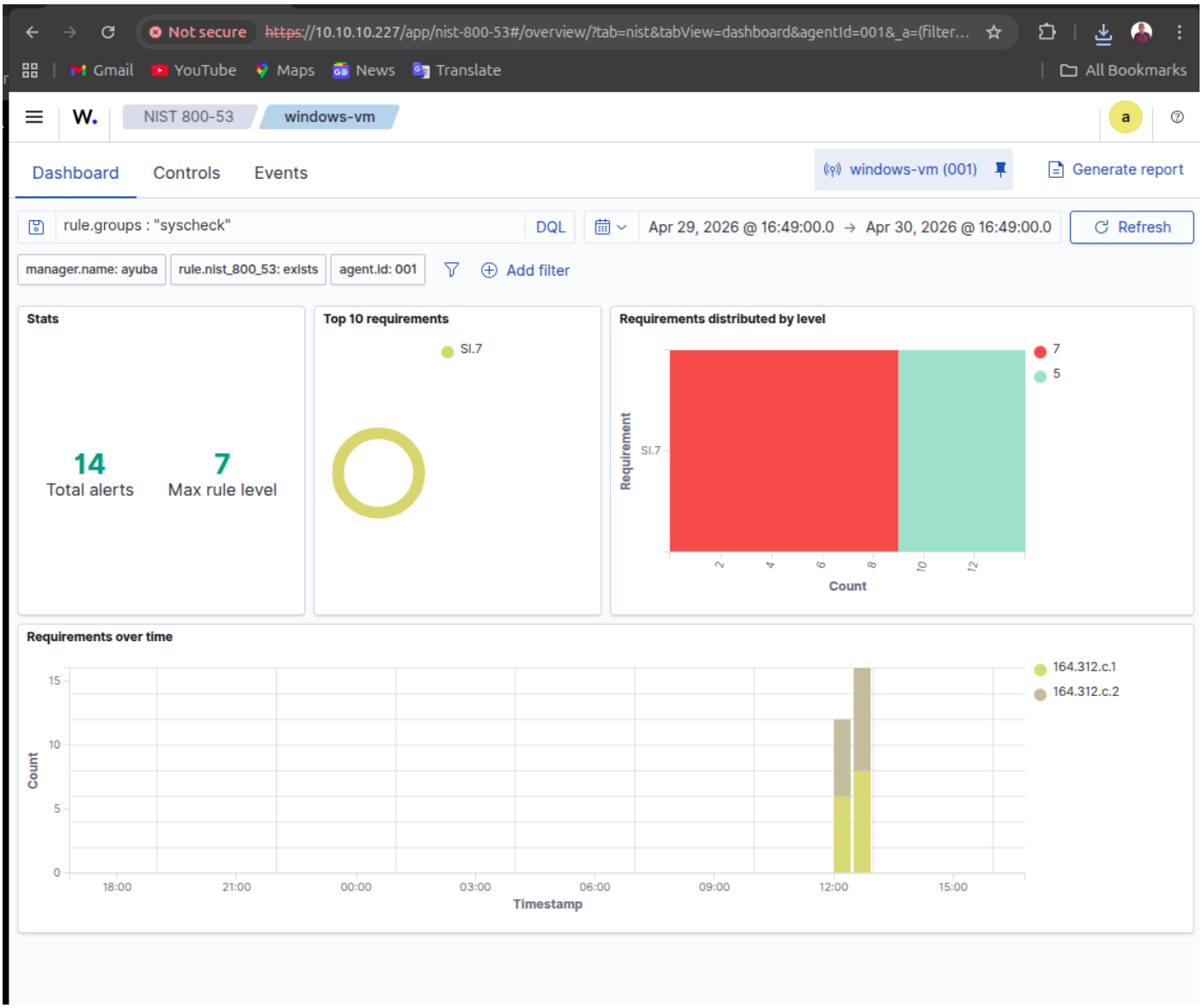
These HIPAA requirements focus on **integrity controls**, ensuring that electronic protected information is not improperly altered or destroyed.

Although this lab used a financial document, the same principle applies to regulated health data. The Wazuh alerts clearly demonstrate that:

- Any modification to a file is detected
- Any deletion is recorded
- Integrity violations can be traced back to a system and time

This aligns directly with HIPAA's requirement for mechanisms that protect data integrity and allow verification of data changes.

9.3 NIST 800-53 Mapping



Observed Mapping

Framework	Control

NIST 800-53	SI-7

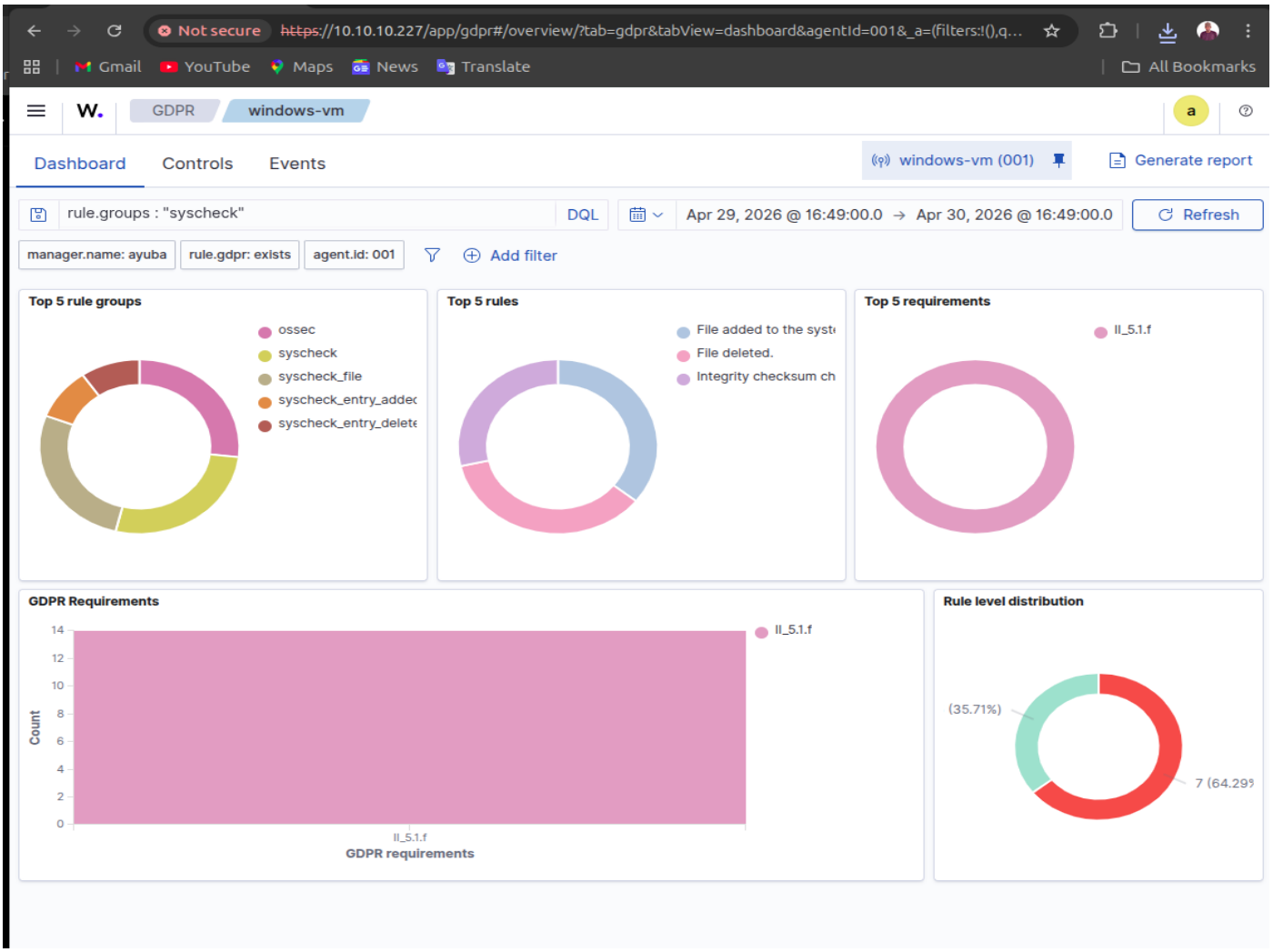
Control SI-7 focuses on **System and Information Integrity**, specifically the detection of unauthorized changes to information and systems.

The lab results demonstrate compliance with this control by showing that:

- File changes trigger integrity alerts
- Hash values are recalculated and compared
- Unauthorized modifications can be identified immediately

This is important because SI-7 is not just about detection, but about **verifiable integrity validation**, which Wazuh provides through checksum comparisons.

9.4 GDPR Mapping



Observed Mapping

Framework	Requirement

GDPR	II_5.1.f

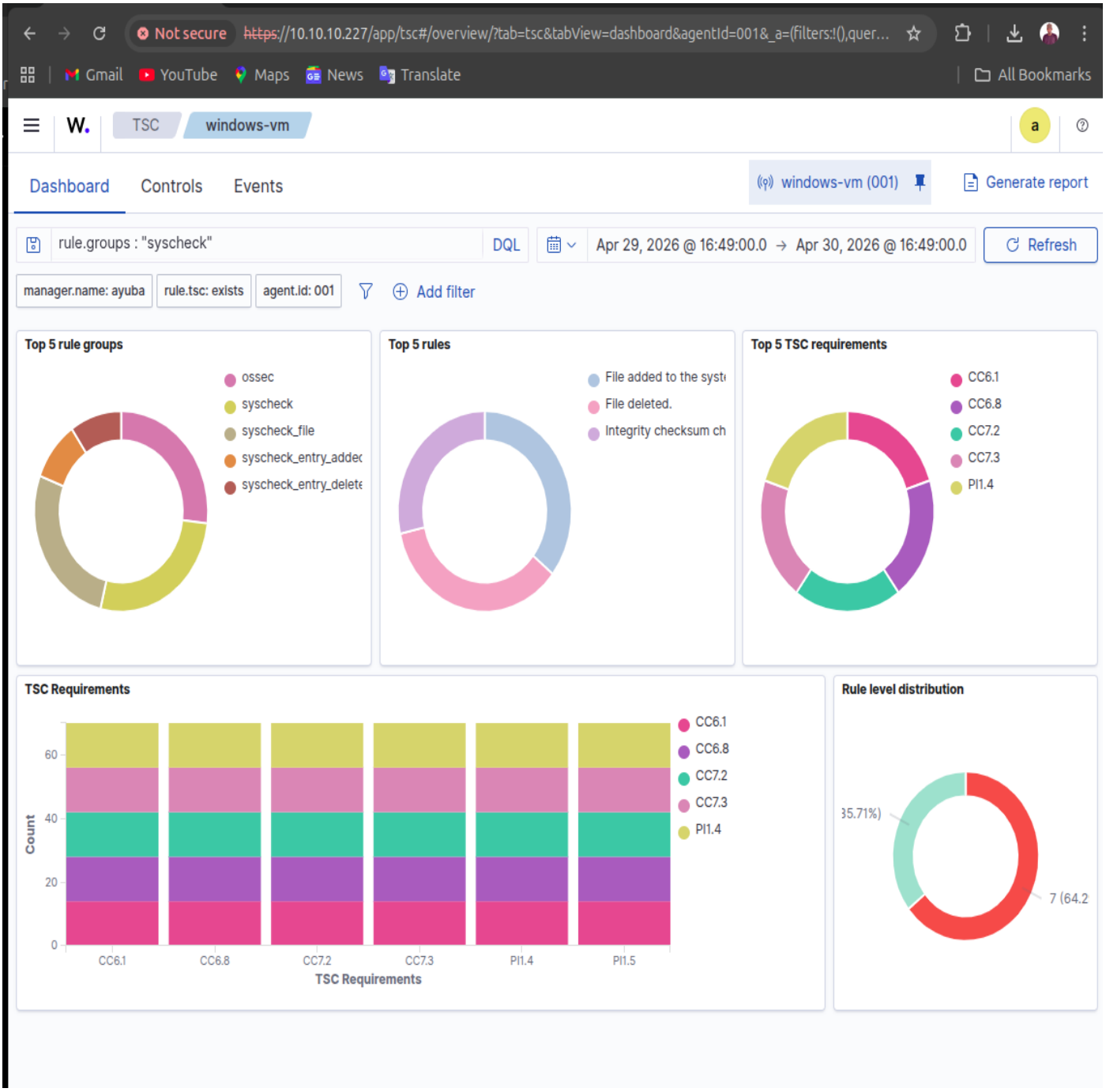
This requirement relates to ensuring the **integrity and confidentiality of personal data**.

Even though the lab focused on a financial document, the same control applies to any file containing personal data. The Wazuh alerts demonstrate that:

- Unauthorized changes to data can be detected
- File integrity is continuously monitored
- Evidence exists to support data protection claims

From a regulatory perspective, this supports the organization's ability to prove that personal data has not been altered without authorization.

9.6 TSC (Trust Services Criteria) Mapping



Observed Mapping

Framework	Requirements
TSC	PI1.4, PI1.5
TSC	CC6.1, CC6.8
TSC	CC7.2, CC7.3

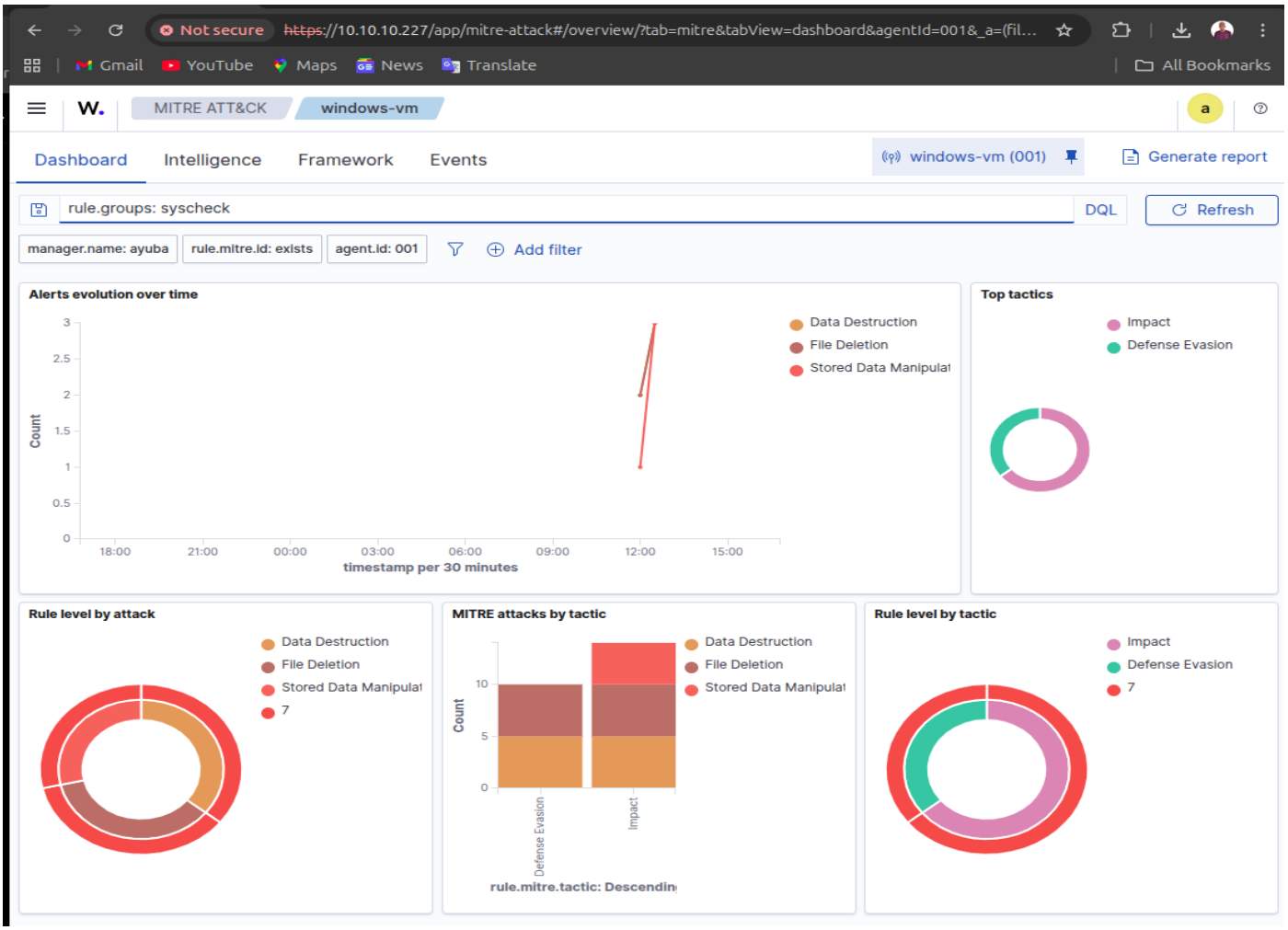
These criteria cover areas such as **processing integrity, logical access, and system monitoring**.

The generated alerts support these requirements in the following ways:

- Detection of file changes supports monitoring controls (CC7.x)
- Tracking user/system interaction with files supports access controls (CC6.x)
- Ensuring data is not altered improperly supports processing integrity (PI1.x)

This shows that the control is not isolated to security alone but contributes to broader governance and assurance objectives.

9.4 MITRE ATT&CK Mapping



Observed Mapping

Category	Mapping
Techniques	File Deletion, Data Destruction, Stored Data Manipulation
Tactics	Defence Evasion, Impact

MITRE ATT&CK is not a compliance framework but a threat behavior model. The mapping shows how the simulated activities align with real attacker techniques.

- File modification aligns with **Stored Data Manipulation**
- File deletion aligns with **Data Destruction**
- These activities fall under **Impact**, where attackers attempt to damage or manipulate data

- The behavior can also relate to **Defense Evasion**, where attackers remove or alter evidence

This mapping strengthens the analysis because it proves that the detected events are not just theoretical—they reflect **real-world attack patterns**.

10. Control Verification Summary

The File Integrity Monitoring control was tested through simulated risk events and validated using Wazuh alerts.

The control consistently detected all expected activities:

Test Activity	Expected Result	Actual Result	Status
File creation	Detection of new file	File added alert generated	Successful
File modification	Detection of integrity change	Checksum change alert generated	Successful
File deletion	Detection of file removal	File deleted alert generated	Successful

11. GRC Workshop and Analysis Questions

11.1 Risk Identification and Analysis

Question 1: What is the potential business impact if unauthorized modification of financial documents occurs?

Unauthorized modification of financial forecast documents can create serious business impact.

The possible impacts include:

Impact Area	Explanation
Financial impact	Incorrect forecasts may lead to wrong budgeting, investment, or spending decisions.
Operational impact	Departments may make decisions based on false or altered data.
Reputational impact	If stakeholders discover that financial records were altered, trust in the organization may reduce.
Compliance impact	The organization may fail audit requirements if it cannot prove document integrity.
Governance impact	Management may lose confidence in internal controls and reporting processes.

Professional Interpretation

Financial forecast documents support business planning and decision-making. If these documents are changed without authorization, leadership may make decisions based on inaccurate information.

This can affect budgets, strategy, reporting, and stakeholder confidence.

Question 2: Based on the alerts generated, how does this control help in quantifying the risk?

The control helps quantify risk by recording how often unauthorized or unexpected changes occur.

From the Wazuh alerts, the organization can measure:

Metric	Purpose
Number of file changes	Measures event frequency
Number of deleted files	Measures possible data destruction activity
Number of modified files	Measures possible tampering attempts
Alert severity	Helps rank risk priority
Affected file paths	Shows which sensitive assets are impacted
Time of event	Supports investigation and timeline analysis

Professional Interpretation

Without monitoring, unauthorized changes may remain invisible. With Wazuh FIM, these events become measurable. This helps the organization identify patterns, measure control effectiveness, and support risk reporting.

11.2 Control Evaluation and Treatment

Question 1: What would be a complementary preventive control for this same risk?

A suitable preventive control would be strict access control on the sensitive folder.

Recommended preventive controls include:

Preventive Control	Purpose
Role-Based Access Control	Ensures only authorized users can access the folder
Least Privilege	Limits users to only the access they need
File permissions	Prevents unauthorized modification or deletion
Multi-Factor Authentication	Reduces unauthorized account access
Backup and versioning	Reduces damage from accidental or malicious deletion

Professional Interpretation

FIM is detective because it identifies changes after they happen. A preventive control should reduce the chance of the change happening in the first place.

The strongest preventive control for this scenario is limiting write and delete permissions to authorized finance personnel only.

Question 2: Explain a situation where the alert from this detective control would trigger a corrective control.

A corrective control would be triggered if Wazuh detects that a sensitive financial forecast file was modified or deleted unexpectedly.

Example response:

1. Wazuh detects that Q4_Financial_Forecast.docx was modified.
2. The security or IT team reviews the alert details.
3. The team checks whether the change was authorized.
4. If unauthorized, the file is restored from backup.
5. The affected user account is reviewed or temporarily disabled.
6. The incident is escalated to management or compliance teams.

Professional Interpretation

The detective control identifies the incident. The corrective control reduces the impact and restores the affected asset.

11.3 Risk Monitoring and Reporting

Question 1: What three key metrics would you pull from the Wazuh dashboard to report operational effectiveness?

The three key metrics are:

Metric	Reason
Number of FIM alerts	Shows how many file integrity events occurred
Alert severity distribution	Shows whether events are low, medium, or high risk
File action type	Shows whether files were added, modified, or deleted

Additional useful metrics include:

Metric	Reason
Affected file paths	Shows which sensitive files were impacted
Agent name	Identifies the affected endpoint
Timestamp	Supports timeline and audit review

Professional Interpretation

These metrics help management understand whether the control is active, whether it is detecting events, and whether the organization is facing repeated file integrity issues.

Question 2: How does centralized monitoring improve governance and auditability?

Centralized monitoring improves governance because all events are collected in one platform.

Without centralized monitoring, the organization would need to check each endpoint manually. This is slow, inconsistent, and difficult to audit.

With Wazuh, the organization can:

Benefit	Explanation
Centralize evidence	All alerts are stored in one dashboard
Improve visibility	Security teams can monitor many endpoints at once
Support audits	Events can be filtered by time, agent, rule, or file path
Improve accountability	Each event is linked to a device and timestamp
Improve reporting	Managers can receive clear metrics on control performance

Professional Interpretation

Centralized monitoring makes risk management more reliable because evidence is easier to collect, review, and report.

11.4 Compliance Mapping

Question 1: How does Wazuh provide evidence for compliance audits?

Wazuh provides audit evidence by recording detailed information about file activity.

The evidence includes:

Evidence Type	Example
Timestamp	When the event occurred
Agent name	Which endpoint generated the event
File path	Which file was affected
Event type	Added, modified, or deleted
Rule description	What Wazuh detected
Hash/checksum	Proof of content change
Framework mapping	PCI DSS, HIPAA, NIST, GDPR, MITRE

Professional Interpretation

This evidence helps auditors verify that monitoring is active and that the organization can detect changes to sensitive files.

Question 2: How would you demonstrate to an auditor that a sensitive file has not been altered during a defined period?

To demonstrate this, I would use the Wazuh dashboard to filter events by:

Filter	Purpose
Agent name	Select the monitored endpoint

File path	Select the sensitive file
Time range	Define the audit period
Rule group	Select syscheck/FIM events

Example:

rule.groups: syscheck AND syscheck.path: C:\SensitiveFiles\Q4_Financial_Forecast.docx

Then I would set the time range to the audit period, such as the last quarter.

Professional Interpretation

If no modification or deletion alerts appear during the selected period, this supports the claim that the file was not altered during that period.

If alerts appear, each alert can be reviewed to determine whether the change was authorized or unauthorized.

General Conclusion

The implementation and testing of the File Integrity Monitoring control demonstrate that the system is operating effectively and meets its intended purpose. The control was able to detect all simulated risk events file creation, modification, and deletion—without failure, providing clear and verifiable evidence for each activity.

This validates that the organization has a functioning mechanism to identify unauthorized changes to sensitive files, which is a critical requirement in both security operations and regulatory compliance. The inclusion of detailed alert data, such as checksums and timestamps, strengthens the reliability of the evidence and supports forensic analysis when needed.

The integration of compliance and framework mappings further elevates the value of the control. It shows that the same technical implementation can simultaneously support multiple regulatory obligations, including data protection, financial security, and system integrity requirements. This reduces duplication of effort and improves overall efficiency in compliance management.

However, the control remains **detective in nature**, meaning it identifies incidents after they occur. For a more complete risk treatment strategy, it should be complemented with preventive controls such as strict access control policies, least privilege enforcement, and user authentication mechanisms. Additionally, corrective controls such as incident response procedures and backup restoration processes should be in place to respond to detected events.

From a governance standpoint, the use of centralized monitoring through Wazuh significantly improves visibility and audit readiness. It allows security teams and management to track events across systems, generate reports, and provide evidence during audits without relying on fragmented or manual log collection.

In conclusion, this work demonstrates a successful alignment between technical security implementation and GRC objectives. The control not only detects and records critical file integrity events but also supports risk analysis, compliance validation, and organizational governance.